L3.4 Linear independence - Part 1

Transcript Language

English

Hello, and welcome to the Maths 2 component of this online B.Sc. program in data science. Today,

we are going to study the topic of Linear Independence. In the last couple of videos,

we have studied the idea of linear dependence. So, in particular, we saw the notion of

linear combinations of vectors. So, let me remind you that now we are studying vectors

in the sense of elements of a vector space, which we defined our two videos before this.

So, we defined the notion of linear combination which is nothing but a some summation aivi, where

vi's are vectors in a vector space and ai's are real numbers. So, this is a finite sum

a1v2 plus a2v2 plus anvn. And we talked about linear dependence.

So, today, in this video, we are going to talk about linear independence.

Let us quickly recall what is linear dependence. So, we say that a set of vectors v1, v2, vn from a

vector space V we call them linearly dependent if there exists scalars a1, a2, an such that

a1v1 plus a2v2 plus anvn is 0. Now, of course, there is a crucial condition extra which is that

not all of the coefficients should be 0. So, at least some coefficient must be non-zero.

If all the coefficients are 0, of course, we know that a1v1 plus a2v2 plus anvn is 0.

So, the point is that there is some linear combination with non-zero coefficients. So,

some coefficients are non-zero so that the sum is still 0. So, if there exist such scalars then we say that v1, v2, vn is linearly dependent. So, equivalently in terms of linear combinations

which is something I defined last time and we just talked about, if we can express the 0 vector as a

linear combination of v1, v2, vn with non-zero coefficients, which means that there are some

coefficients, meaning at least one coefficient which is non-zero, then we say that v1,

v2, vn are linearly dependent. So, we have studied this notion in the previous video.

So, in this video, we are going to talk about linear independence. So, we say that a set of

vectors v1, v2, vn from a vector space V we call them linearly independent if they are

not linearly dependent. So, this is a very easy definition. We have already studied what is linear

dependence. So, we will say they are linearly independent if they are not linearly dependent.

So, what does this mean in terms of linear combinations and summing up to 0?

So, the equivalent formulation is a set of vectors v1, v2, vn from a vector space V is said to be linearly independent if the equation a1v1 plus a2v2 plus anvn is 0

can only be satisfied when ai is 0. So, just to go back to dependence, dependence said there is

some linear combination, there is some, there are some scalars a1, a2, an, not all of which are 0,

so that when you take the sum a1v1 plus a2v2 plus anvn you get 0. Linear independence is saying,

if there is a linear combination, meaning if there is a sum a1v1 plus a2v2 plus anvn which is 0,

then the only way in which this can happen is if the ai's are all 0.

So, equivalently in terms of linear combinations, we are saying that a set of vectors v1, v2,

vn is linearly independent if the only linear combination

of these vectors v1, v2, vn which equals 0 is the linear combination with all coefficients 0.

So, let us take a pause and understand what we are saying here. If the coefficients are 0, certainly

summation aivi is 0. If all the ai's are 0, then summation aivi is going to be 0.

So, what we are saying is that if the sum is 0, then the coefficients must be 0. In other words, the only way of getting 0 as a linear combination is if the coefficients are 0.

So, I want to emphasize that this means we have to check something is linearly independent,

which we will do at the end of this video. We have to check that all the ai's are 0.

So, let us do some examples. So, let us first check out this example in R2.

So, look at the two vectors minus 1 comma 3 and 2 comma 0.

So, we want to see if which coefficients give us that the sum is 0. So, you write this

equation, a times minus 1 comma 3 plus b times 2 comma 0 is 0 comma 0 and then you

try to solve for a and b. So, if you do that, you equate the corresponding coordinates.

So, the first coordinate for the left hand side is minus a plus 2 times b, the first coordinate for

the right hand side is 0. They are equal so that means these two must be the same. So, minus a plus

2b is 0. And the second coordinate is 3 times a plus 0 times b, which is just 3 times a. And the

second coordinate on the right is 0. So, you get 3 times a is 0. So, we have a system of linear equations, minus a plus 2b is 0 and 3a is 0. Now, we know how to solve a system of linear

equations. We in fact know very general method for that. And indeed, towards the end of the video,

we will start using that method. But for now, you can see this is very easy to solve. Namely,

3 times a 0, so a must be 0. And once a is 0, you put that into the first equation

and you get that 2 times b is 0. So, b is 0. So, the unique solution for this system is a 0

and b 0, which means that if a times minus 1 comma 3 plus b times 2 comma 0 is 0 comma 0,

the only solutions of, the only coefficients which yield this identity to be true are a is 0 and b is

0. That means, the vectors minus 1 comma 3 and 2 comma 0 are linearly independent.

Let us look at the 0 vector. This is a very special vector in our vector space. So, suppose

I have a set of vectors v1, v2, vn and one of the vectors here is the 0 vector. So, suppose vi is 0.

One of them is 0. Let us say that one is vi. So, then what you do is you choose the ith coordinate, sorry, you choose the ith coefficient to be 1 and you choose the jth coefficient, where j is not i

to be 0. So, if you do that, then look at this linear combination, a1v1 plus a2v2 plus anvn,

now for all the vectors where j is not i, aj times vj is going to be 0, because aj is 0.

On the other hand for the ith vector, you have ai times vi, vi is 0, so ai times vi is also 0. So,

for each of these terms, a1v1, a2v2 etc. up till anvn each term is 0. Hence, the sum is also 0.

So, the linear combination is 0. But of course, all the coefficients are not 0, because ai is 1.

So, one of the coefficients is 1. So, not all these coefficients are 0. So, what does that mean?

That means a set of vectors which contains the 0 vector is always a linearly dependent set. So,

this is not linearly independent. So, if your v1, v2, vn happens to contain 0, then

this is linearly dependent. It is not linearly independent. So, let us keep this in mind.

So, now let us ask the question when a two non-zero vectors linearly independent. So, we have already seen that if a vector is 0, you take that in your collection of vectors,

then that is a linearly dependent set. So, now we can take two non-zero vectors. That is your first

starting point for our discussion. So, let us take two non-zero vectors and ask

when are they linearly independent? When is the set of vectors linearly independent?

So, let v1 and v2 be two non-zero vectors. So, suppose v1 and v2 are linearly dependent.

So, we want to study when they are linearly independent instead we will study when they are

linearly dependent and from there we will conclude when they are linearly independent. So, suppose v1

and v2 are linearly dependent that means a1v1 plus a2v2 is 0 for some coefficients a1 and a2,

and these coefficients, both of these coefficients are not 0. So, we are saying a1v1 plus

a2v2 is 0 for some coefficients a1 and a2,

where both of them, I mean, at least one of them is non-zero. So, at least one

of a1 or a2 is not 0. That is the definition of a linear dependence.

But the point is here, since the vectors are non-zero, if one of them, one of these

coefficients is non-zero, then the other better be non-zero, because otherwise you will have,

let us say, a1 is non-zero, but a2 is 0, then you will have a1 times v1 which is a non-zero vector

equates to 0 that cannot be possible. So, if one of them is not zero, then both of them are not

0. And really, that is what this statement here means. So, keep in mind that one of a1 or a2 is

non-zero and so both of them must be non-zero. That was the import of this statement here.

So, dividing by a1 and putting c is minus a2 by a1, we get that v1 is c times v2. And so v1 and

v2 are multiples of each other. So, what are we saying? We are saying that if v1 and v2 are

linearly dependent, then v1 and v2 are multiples of each other. And we can go backwards. We can

reverse this set of statements. So, we can reverse the implications above and conclude that if v1

and v2 are multiples of each other, then they are linearly independent. If v1 and v2 are multiples,

then v1 is c times v2 for some c. And then note that c is non-zero, because both v1 and v2 are non-zero vectors and then you can write this as v1 minus c times v2

is 0 and so the coefficients are 1 and minus c, and in fact, both of them are non-zero.

Actually, it is enough for them, one of them to be non-zero. But in this case,

both of them are non-zero. And so we get that they are linearly dependent. So, that is what I mean by

we can reverse the implications. So, the conclusion is, if two,

you have two non-zero vectors, then they are linearly independent precisely when they are not

multiples of each other. We have seen that linear dependence of two non-zero vectors is equivalent

to them being multiples of each other. So, linear independence is exactly same as saying that they

are not multiples of each other. Let me qualify this coefficient a1 and a2 so, where at least

one of a1 or a2 is not 0. And as I said, because both the vectors are non-zero and you have only

two vectors in this case both of them must be non-zero. That is what we are saying.

So, the take home from here is, if you have two non-zero vectors, then linear independence means that they are not multiples of each other. So, the point one is trying to make here is that

linear independence as a notion is a very useful and important notion and it is saying something

very important about these vectors, namely that they are not multiples.

So, let us ask the same of what happens to three vectors? Let us ask the same question

for three vectors. When the three vectors linearly independent what does that say? So, we will do the

same thing. We will first study when they are dependent and then from there we will try to

draw conclusions for when they are independent. So, suppose v1, v2 and v3 are linearly dependent

then we have an equation of the form a1v1 plus a2v2 plus a3v3 is 0 where for some

coefficients a1, a2, a3 where at least one of the coefficients is non-zero, which is exactly

what I wanted to say on the previous slide as well. At least one of these is non-zero.

So, let us assume that a1 is non-zero. We will, in a minute we will also study the other cases, but this is a prototype case. Let us study a1 is not 0. So, if a1 is not 0, I can divide by a1

and I can take the terms for v2 and v3 on the other side and I can write v1 as b2v2

plus b2v2, b3v3, this should be b3v3, where b2 is minus a2 by a1 and b3 is minus a3 by a1.

So, v1 is a linear combination of the other two vectors that is the main point. This is b3.

So, now we can make the same argument of a2 is non-zero or a3 is non-zero. Remember that we

know that one of them is non-zero. If it is not a1, maybe a2 is non-zero, maybe a3 is non-zero.

And you can see that you can make the same argument and express. So, if a2 is non-zero,

you can express v2 as a linear combination of the other two. And if a3 is non-zero, you can

express v3 as a linear combination of the other two vectors. And I again note that this is b3.

So, once again these implications are reversible. So, if, let us say

if v1 is b2v2 plus b3v3, so then I can write, I can take the v2 and v3 terms on the other side and

write v1 minus b2v2 minus b3v3 is 0, but remember that v1 has coefficient 1. So, I get 1 times v1 minus b2 times v2 minus b3 times v3 is 0, which tells me that I have a linear combination where at

least one of the coefficients is non-zero, which one, the coefficient for v1, because that is 1.

And so this is linear, they are linearly dependent. And you can argue the same way

if v2 is a linear combination of v1 and v3 or v3 is a linear combination of v1 and v2.

So, these implications are reversible. So, the upshot is that,

if you have three vectors v1, v2, v3, which are linearly dependent, this is exactly the

same as saying that one of these vectors is a linear combination of the other two vectors.

So, now we can talk about linear independence. We have studied when they are linearly dependent.

So, we can now answer the question about linearly independent.

So, if three vectors are linearly independent, then none of these vectors

is a linear combination of the other two. That is what we are seeing.

So, you can think of this geometrically. I will, we will do that in a few minutes.

But already you can start thinking of what this means in terms of geometry.

So, the conclusion here is the take home. If three vectors are linearly independent,

this is exactly the same as saying that none of them is a linear combination of the other two.

Let us look at an example in R3. So, let us take these three vectors 1, 1, 2, 1, 2, 0 and 0, 2,

1. So, we want to ask whether or not they are linearly independent. So, let us take some

arbitrary coefficients a, b and c. So, a times 1, 1, 2 plus b times 1, 2, 0 plus c times 0, 2, 1 is

0, 0, 0. So, we equate this to 0, 0, 0. And then we ask, can we get from here what are the choices

for a, b, c, in particular, is there a choice where at least one of a, b or c is non-zero.

So, we have the following system of linear equations. How do we get this? By equating the

corresponding coordinates. So, the first coordinate on the left hand side is

a times 1 plus b times 1 plus c times 0, which is a plus b.

The first coordinate on the right hand side is 0, so this is a plus b is 0. Similarly, if you equate the second coordinates, you get a plus 2b plus 2c is 0. And the third coordinate then you

get 2a plus 0 times b plus 1 times c is 0. So, that is how you get the system of equations.

Now, let us solve for a, b and c and see if we can get solutions which are non-zero.

So, from the first equation we get that b is minus a, from the third equation we get c is minus 2a,

if you substitute into the middle equation, you get some combination of a is

not 0. So, specifically, you get b, a minus 2 times a minus 4 times a. So, a minus 6 times a,

so minus 5a is 0, which tells you that a is 0. And then once a is 0, b is minus a, so b is 0,

and c is minus 2a, so c is 0. So, the only equation for this system is a is, b is, c is 0.

So, what have we achieved? We have achieved that if you have an equation like this with

coefficients a, b, and c, then the only way this equation is satisfied, when I say this equation,

I mean, this equation here, the only way this equation is satisfied is when a is 0, b is 0 and c

0. That tells us that these three vectors 1, 1, 2, 1, 2, 0 and 0, 2, 1 are linearly independent.

So, we have seen in this video the notion of linearly, a linearly independent set of vectors.

So, we say that a set of vectors is linearly independent if they are not linearly dependent,

which really means that if you take a linear combination of these vectors and equate it to 0,

the only way that that is possible is if all the coefficient are 0. Thank you.